Analytical Cumulative Examination
October 25, 1997

1. (40) A laser pointer produces 5 mW at 630 nm with a 1 nm FWHM in a beam 1 mm^2.
   a. Calculate the number of photons s^{-1} mm^{-2} nm^{-1} produced by the laser pointer.
   b. A tungsten filament can be approximated as a perfect black body radiator. Calculate the temperature of a tungsten filament needed to produce a maximum wavelength of emission at 630 nm.
   c. The melting point of tungsten is 3410 °C. Determine the spectral radiance at 630 nm (B_λ) for a tungsten filament at its melting point (color temperature = 3410 °C).
   d. Convert B_λ to units of photons s^{-1} sr^{-1} mm^{-2} nm^{-1} and compare this answer to that in a.

$$B_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1} \qquad \lambda_{\max} = \frac{hc/k}{4.9651T}$$

Use order-of-magnitude values for the physical constants.

2. (30) Using the data below, calculate the following parameters.

Standard Conc (M)	Analyte Response	Blank Response	Std Dev. Blank
1×10^{-5}	1.90	0.4	0.07
5×10^{-6}	1.10	0.4	0.06
2×10^{-6}	0.75	0.5	0.05
1×10^{-6}	0.43	0.3	0.05

   a. Sensitivity.
   b. Signal to noise ratio (SNR) for lowest concentration standard.
   c. Limit of detection (in M) at a SNR =2.
   d. The number of samples was increased by a factor of 10 and averaged together. By what factor would the SNR improve.

3. (30) An experiment is changed such that the signal is increased by a factor of 2 and the background is increased by a factor of 4. **Quantitatively** determine the effect on the signal to noise ratio if the following noise sources limit the experiment. Explain your answers.

   a. detector dark current
   b. background proportional (flicker) noise
   c. signal shot noise
   d. background shot noise
   e. signal proportional noise